

CHAPTER 4

ANALYZING FUNCTIONAL DEPENDENCIES, REDUNDANCY AND DATA ANOMALIES

4.1 Introduction

The StyleNova E-Commerce Database System is designed to efficiently manage customers, sellers, products, orders, payments, inventory, and product reviews. Functional dependencies define the relationship between attributes within each table, while redundancy analysis and anomaly detection help ensure data consistency, reduce duplication, and maintain database integrity. A well-designed relational database improves performance and supports reliable e-commerce operations.

1. CUSTOMER TABLE

Functional Dependency

Customer_ID → First_Name, Last_Name, Email, Phone, Address, Password

Redundancy Analysis

Customer details are stored only once in the Customer table. Other tables use only Customer_ID as a foreign key, preventing duplicate customer information.

2. SELLER TABLE

Functional Dependency

Seller_ID → Seller_Name, Business_Name, Email, Phone, Address

Redundancy Analysis

Seller information is maintained only in the Seller table. Products reference sellers using Seller_ID, avoiding repeated seller details.

3. PRODUCT TABLE

Functional Dependency

Product_ID → Product_Name, Category, Price, Stock, Description, Seller_ID

Redundancy Analysis

Product information is stored only once in the Product table. Orders and reviews reference products using Product_ID, minimizing data duplication.

4. ORDER TABLE

Functional Dependency

Order_ID → Customer_ID, Order_Date, Order_Status, Total_Amount

Redundancy Analysis

The Order table stores only order-related details. Customer information is accessed using Customer_ID, avoiding repeated customer data.

5. PAYMENT TABLE

Functional Dependency

Payment_ID → Order_ID, Payment_Method, Payment_Status, Payment_Date, Amount

Redundancy Analysis

Payment details are stored separately from the Order table, preventing duplicate payment records.

6. INVENTORY TABLE

Functional Dependency

Inventory_ID → Product_ID, Quantity_Available, Last_Updated

Redundancy Analysis

Inventory records are maintained separately, ensuring stock information is not repeatedly stored in the Product table.

7. REVIEW TABLE

Functional Dependency

Review_ID → Customer_ID, Product_ID, Rating, Review_Text, Review_Date

Redundancy Analysis

Reviews are stored in a dedicated table. Customer and product details are referenced through foreign keys, avoiding repeated information.

CONCLUSION

The StyleNova E-Commerce Database System is designed using appropriate functional dependencies and normalized relational tables. Each entity stores only its relevant information, reducing redundancy and preventing insert, update, and delete anomalies. This design improves data consistency, maintains referential integrity, and supports efficient management of customer accounts, products, sellers, orders, inventory, payments, and reviews in the StyleNova platform.

